Отчёт по лабораторной работе 1

Введение в mininet.

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Содержание

1 Цель работы

Основной целью работы является развёртывание в системе виртуализации (например, в VirtualBox) mininet, знакомство с основными командами для работы с Mininet через командную строку и через графический интерфейс.

2 Ход работы

Запускаем машину mininet и хостовую виртуальную. Заходим в mininet с хостовой.

```
mininet@mininet-vm: ~
          [-O ctl_cmd] [-o option] [-P tag] [-p port] [-R address]
          [-S ctl_path] [-W host:port] [-w local_tun[:remote_tun]]
          destination [command [argument ...]]
       ssh [-Q query_option]
sacvelev@sacvelev:~$ ssh -Y mininet@192.168.56.102
The authenticity of host '192.168.56.102 (192.168.56.102)' can't be established.
ED25519 key fingerprint is SHA256:ZSNhVBd3TMcF4yf5qNCx9YVEM/pVSBb8PMB1dnGwh5o.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.102' (ED25519) to the list of known hosts
.
mininet@192.168.56.102's password:
Welcome to Ubuntu 20.04.1 LTS (GNU/Linux 5.4.0-42-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your
Internet connection or proxy settings

Last login: Tue Feb 10 18:43:33 2026
/usr/bin/xauth:  file /home/mininet/.Xauthority does not exist
mininet@mininet-vm:~$
```

Смотрим информацию о соединении mininet. Делаем так, чтобы оно имело адрес вида 1.0.0.x.x

```
mininet@mininet-vm: ~
        inet 192.168.56.102  netmask 255.255.255.0  broadcast 192.168.56.255
        ether 08:00:27:7f:7e:6d  txqueuelen 1000  (Ethernet)
        RX packets 155  bytes 22183 (22.1 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 91  bytes 15009 (15.0 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        ether 08:00:27:01:d9:22  txqueuelen 1000  (Ethernet)
        RX packets 2  bytes 1180 (1.1 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 2  bytes 684 (684.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 1752  bytes 134248 (134.2 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 1752  bytes 134248 (134.2 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

Позволяем mininet использовать два адаптера при запуске. Скачиваем новую версию Mininet. И проверяем версию.

```
Cloning into 'mininet'...
remote: Enumerating objects: 10388, done.
remote: Counting objects: 100% (131/131), done.
remote: Compressing objects: 100% (60/60), done.
remote: Total 10388 (delta 104), reused 71 (delta 71), pack-reused 10257 (from 3)
Receiving objects: 100% (10388/10388), 3.36 MiB | 5.09 MiB/s, done.
Resolving deltas: 100% (6906/6906), done.
mininet@mininet-vm:~$ cd ~/mininet
mininet@mininet-vm:~/mininet$ sudo make install
cc -Wall -Wextra  \
-DVERSION=\"`PYTHONPATH=. python -B bin/mn --version 2>&1`\" mnexec.c -o mnexec
install -D mnexec /usr/bin/mnexec
PYTHONPATH=. help2man -N -n "create a Mininet network." \
--no-discard-stderr "python -B bin/mn" -o mn.1
help2man -N -n "execution utility for Mininet." \
-h "-h" -v "-v" --no-discard-stderr ./mnexec -o mnexec.1
install -D -t /usr/share/man/man1 mn.1 mnexec.1
python -m pip uninstall -y mininet || true
Found existing installation: mininet 2.3.0
Uninstalling mininet-2.3.0:
  Successfully uninstalled mininet-2.3.0
python -m pip install .
Processing /home/mininet/mininet
Requirement already satisfied: setuptools in /usr/lib/python3/dist-packages (from mininet==2.3.1b4)
(45.2.0)
Building wheels for collected packages: mininet
  Building wheel for mininet (setup.py) ... done
  Created wheel for mininet: filename=mininet-2.3.1b4-py3-none-any.whl size=160942 sha256=ec95bc3037
486a2fbdf5d71d63adcbd8a806fe8588bbcf8bcd9da67f37310727
  Stored in directory: /tmp/pip-ephem-wheel-cache-72fjjggt/wheels/cd/7d/a7/aafe1b3eaff31efd6ba4e2ea6
c9690a717bdf739db6cfe8d45
Successfully built mininet
Installing collected packages: mininet
Successfully installed mininet-2.3.1b4
mininet@mininet-vm:~/mininet$ mn --version
2.3.1b4
```

Далее, позволяем приложениям запускаться от пользователя mininet.

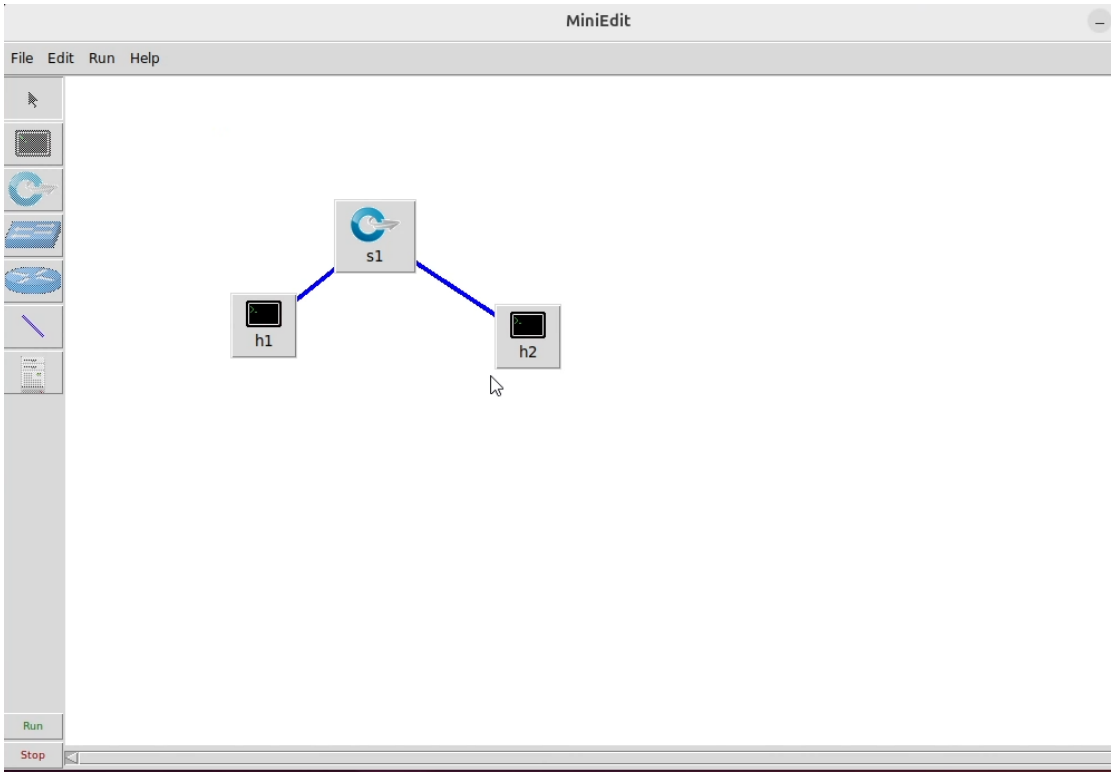
```
root@mininet-vm:~# xauth list
xauth:  file /root/.Xauthority does not exist
root@mininet-vm:~# ^C
root@mininet-vm:~# xauth add mininet-vm/unix:10 MIT-MAGIC-COOKIE-1 87b6ea35aad8f
96fc2db94178421d3bd
xauth:  file /root/.Xauthority does not exist
root@mininet-vm:~# xauth add mininet-vm/unix:10 MIT-MAGIC-COOKIE-1 87b6ea35aad8f
96fc2db94178421d3bd
root@mininet-vm:~# xauth list $DISPLAY
mininet-vm/unix:10  MIT-MAGIC-COOKIE-1  87b6ea35aad8f96fc2db94178421d3bd
root@mininet-vm:~# logout
```

Создаём минимальную топологию и смотрим её параметры.

```
s1
*** Adding links:
(h1, s1) (h2, s1)
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet> help

Documented commands (type help <topic>):
========================================
EOF    gterm  iperfudp  nodes        pingpair      py      switch  xterm
dpctl  help   link      noecho       pingpairfull  quit    time
dump   intfs  links     pingall      ports         sh      wait
exit   iperf  net       pingallfull  px            source  x

You may also send a command to a node using:
  <node> command {args}
For example:
  mininet> h1 ifconfig

The interpreter automatically substitutes IP addresses
for node names when a node is the first arg, so commands
like
  mininet> h2 ping h3
should work.

Some character-oriented interactive commands require
noecho:
  mininet> noecho h2 vi foo.py
However, starting up an xterm/gterm is generally better:
  mininet> xterm h2

mininet>
```

```
        ether 08:00:27:7f:7e:6d  txqueuelen 1000  (Ethernet)
        RX packets 1398  bytes 135154 (135.1 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 924  bytes 178795 (178.7 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        ether 08:00:27:01:d9:22  txqueuelen 1000  (Ethernet)
        RX packets 4053  bytes 5866427 (5.8 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 899  bytes 60600 (60.6 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 1835  bytes 140158 (140.1 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 1835  bytes 140158 (140.1 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

s1-eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        ether 92:dd:5f:67:f6:54  txqueuelen 1000  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

s1-eth2: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        ether 52:8f:ba:8a:73:c6  txqueuelen 1000  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

mininet> _
```

Создаём в MiniEdit топологию. Пингуем хосты, проверяем работоспособность.



3 Вывод

Мы изучили как работать с mininet.